FEDERAL SPECIFICATION

AMBULANCE

Emergency Medical Care Vehicle

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers new commercially produced surface emergency medical care vehicles, hereinafter referred to as ambulance, or vehicle. The ambulance is self-propelled, front or rear wheel driven (4 x 2) or four wheel driven (4 x 4); designed, constructed, and equipped to afford maximum safety and comfort to patients and personnel (for intended use, see section 6). Vehicles procured under this specification shall be warranted by the manufacturer or contractor as specified in the procurement document.

1.2 Classification.

1.2.1 Ambulance types, classes, and floor plans. Ambulances covered by this specification shall be of the following, as specified (see 6.2):

Type I - Conventional, cab-chassis with modular ambulance body(3.1.2 and figure 1).

Class 1 - Two rear wheel driven (4 x 2).

Class 2 - Four wheel driven (4 x 4).

Floor plan A - Elevating cot and squad bench(3.1.5; 3.11.4) standard.

Floor plan B - Elevating cot and wheeled cot-bench(3.1.5; 3.11.5) optional.

Type II - Standard van, forward control (FC) integral cab-body ambulance(3.1.3 and figure 2).

Class 1 - Two rear wheel driven (4 x 2).

Class 2 - (Not yet available).

Floor plan A - Elevating cot and squad bench(3.1.5; 3.11.4) standard.

Floor plan B - Elevating cot and wheeled cot-bench(3.1.5; 3.11.5) optional.

Type III - Specialty van, forward control (FC) integral cab-body ambulance (3.1.4 and figure 3).

Class 1 - Two rear wheel driven (4 x 2).

Class 2 - (not yet available).

Class 3 - Two front wheel driven (4 x 2).

Floor plan A - Elevating cot and squad bench(3.1.5; 3.11.4) standard.

Floor plan B - Elevating cot and wheeled cot-bench(3.1.5; 3.11.5) optioanl.

1.3 Standard ambulance. A vehicle in compliance with this specification shall be defined as a standard ambulance. This ambulance is in general accord with the Ambulance Design Criteria of the National Highway Traffic Safety Administration, U.S. Department of Transportation, Washington, DC, as reported by the Committee on Ambulance Design Criteria, National Research Council, National Academy of Engineering, Washington, DC.

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1.4 Ambulance parameters and options. The requirements as specified hereinafter shall apply to all ambulances covered in this specification unless otherwise shown, annotated, or amended by an authorized directive. Additional options, or alternates may be specified by following the ordering data procedures (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

W-B-00131J - Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
L-S-300 - Sheeting and Tape, Reflective, Nonexposed Lens, Adhesive Backing.
RR-C-901 - Cylinders, Compressed Gas: With Valve or Plug and Cap; ICC3aa.
ZZ-I-550 - Inner Tube Pneumatic Tire.

Federal Standards:

Federal Std. No. 297 - Rustproofing of Automotive Vehicles.

Fed. Std. No. 595 - Colors Vol. I.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-1223 - Administrative Wheeled Vehicles Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.

MIL-STD-39226 - Cylinder Compressed Gas, DOT spec. 3AA2015, Medical Gases.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws, Rules and Regulations:

Environmental Protection Agency:

Title 40 CFR, Part 85 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

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U.S. Department of Transportation:

National Highway Traffic Safety Administration
49 CFR 390 - Federal Motor Vehicle Safety Standards (FMVSS)

Federal Highway Administration
49 CFR 571 - Motor Carrier Safety Regulations (MCSR)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Tire and Rim Association Inc. Yearbook

Application for copies should be address to The Tire and Rim Association, Inc. 3200 West Market Street, Akron, Ohio 44313

Society of Automotive Engineers (SAE), Inc., Standards and Recommended Practices:

J245 - Engine Rating Code, Spark Ignition.

J551a - Measurement of Electromagnetic Radiation from Motor Vehicles (20-1,000MHz).

J552a - External Electromagnetic Radiation Suppressors.

J579a - Seal Beam Headlamp units for Motor Vehicles.

J595b - Flashing Warning Lamps for Authorized Emergency, Maintenance and Service Vehicles.

J598a - Sealed Lighting Units for Construction and Industrial Machinery.

J683a - Tire Chain Clearance.

J689a - Approach, Departure, and Ramp Breakover Angles.

J695a - Turning Ability and Off Tracking.

J845 - Emergency Warning Lamp. 360 Deg.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

Static Load Test Code for School Bus Body Structure:

Standard No. 002.

(Application for copies should be addressed to the School Bus Manufacturers Institute, A Division of Truck Body and Equipment Association, Inc., 5530 Wisconsin Avenue, N.W., Washington, DC 20015.)

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International Mobile Air Conditioning Association, Inc.:

IMACA Standard 200

(Application for copies should be addressed to the International Mobile Air Conditioning Association, Inc., 6116 North Central Expressway, Dallas 6, Texas.)

Electronic Industries Association, Standards:

RS152B - Minimum Standards for Land Mobile Communication FM or PM Transmitters.

RS204A - Minimum Standard for Land Mobile Communication FM or PM Receivers.

RS220 - Continuous Tone Controlled Squelch Systems.

RS329 - Minimum Standard for Land Mobile Communication Antennas Part II Mobile Antennas.

RS374 - Land Mobile Selective Signalling Standard.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W. Washington, DC 20006.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 General vehicular design, types, and floor plan.

3.1.1 Design. The ambulance and the allied equipment furnished under this specification shall be the manufacturer's current commercial vehicle of the type and class specified. The ambulance shall be complete with the operating accessories as specified herein; furnished with such modifications and attachments as may be necessary and specified to enable the vehicle to function reliably and efficiently in sustained operation. The design of the vehicle and the specified equipment shall permit accessibility for servicing, replacement, and adjustment of component parts and accessories with minimum disturbance to other components and systems. The term "heavy-duty" as used to describe an item, shall mean in excess of the usual quantity, quality, or capacity that is normally supplied with the standard production vehicle, or component.

3.1.2 Type I ambulance. Type I vehicle, class 1 or 2 as specified, shall be a chassis furnished with a 2-door conventional cab. Chassis-cab shall be suitable for subsequent mounting of a modular (containerized) transferrable equipped ambulance body conforming to the requirements specified herein.

3.1.3 Type II ambulance. Type II vehicle, class 1, forward control (FC) with engine compartment located between driver and assistant, shall be an integral cab-body standard commercial compact van of unitized construction or on a frame constructed body requiring only modifications in height, and equipped as specified herein.

3.1.4 Type III ambulance. Type III vehicle, class 1 or 3 as specified, forward control (FC) shall be a cab-body of unitized construction or on a frame constructed integral ambulance unit. The van chassis, van forward section chassis, or motor home chassis furnished shall be suitable for the subsequent fabrication, conversion or modification into a single unit van ambulance incorporating and conforming to the requirements specified herein.

3.1.5 Floor plan of patient-technician compartment. Unless otherwise specified (see 6.2), floor plan "A" loading arrangement shall be provided in the patient's compartment ambulance body proper. All litters shall be loaded with the heads of the patients forward in the vehicle. The following plan of patient-technician potential loading accommodations shall be furnished:

Floor plan A - One patient (primary patient) on a wheeled elevating cot and one (secondary patient) lying on a folding stretcher or combination stretcher chair on the squad bench platform; or one primary patient and three secondary seated patients on the squad bench (see 3.11.4), and one seated technician (see 3.10.3).

Floor plan B - One patient (primary patient) on a wheeled elevating cot and one (secondary patient) lying on a wheeled non-elevating bench type cot capable of accommodating three seated patients (see cots 3.11.5); and one seated technician (see 3.10.3).

NOTE: Primary patient must be positioned on the left side of the ambulance. The anatomy of the tracheal bronchial tree is so constructed that the unconscious patient will be less likely to aspirate fluids into his trachea if he is placed on his left side with his right side up.

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3.2 Vehicle, ambulance components, equipment and accessories. The emergency medical care vehicle, chassis, ambulance bodies, equipment, devices, medical accessories and electronic equipment to be delivered under this contract shall be standard commercial products, which meet or exceed the requirements of this specification. The ambulance shall comply with all Federal Motor Vehicle Safety Standards, (FMVSS) and Federal Regulations applicable or specified for the year of manufacture. The chassis, components, and optional items shall be as represented in the manufacturer's current technical data, and the ambulance bodies, equipment, and accessories shall be as represented in their respective manufacturer's current technical data. Data shall be limited to specifications and technical materials, identical to that furnished to the authorized company representatives, and shall be on file in appropriate offices of the procuring activity prior to delivery of the item. The ambulance components and equipment need not be the products of the same manufacturer's. The supplier shall provide total standardization and inter-changeability between similar vehicles and alike equipment, items, and accessories specified.

3.3 Materials. The materials shall be new and not less than the quality conforming to current engineering and manufacturing practices. Materials shall be free of defects, and suitable for the intended service.

3.4 Vehicle operation, performance and physical characteristics.

3.4.1 Operation and performance. All requirements as specified in section 3, are minimums and shall be met with the ambulance laden to the Gross Vehicle Weight Rating, and all specified equipment and devices installed and operating at maximum power-consuming condition; i.e., air conditioning, lights, and radio(s). The vehicle shall be capable of operating safely and efficiently under environmental conditions outlined herein or as specified in the invitation for bid, contract, or order.

3.4.2 Temperature conditions. The ambulance, equipment and devices thereon, shall be capable of normal satisfactory performance in ambient temperatures of -30° to 125°F., when serviced in accordance with the manufacturer's recommendation.

3.4.3 Noise and sound level limits, exterior. Unless more stringent sound levels are regulated by the states and municipalities where the ambulance will be based, the exterior noise level produced by the vehicle shall not exceed Federal regulations.

3.4.4 Vehicle performance. The ambulance shall assure a smooth, stable, ride, with nominal noise and vibration; with due consideration of road conditions. The ambulance shall be capable of rapid response, and accelerate to speeds consistent with street traffic and interstate highways.

3.4.5 Brakes. The ambulance as delivered to the user shall comply to performance values required by Federal Motor Vehicle Safety Standards (FMVSS) 106, 116, this specification, and FMVSS 105 when applicable.

3.4.6 Speed. Vehicle shall be capable of a sustained speed of not less than 70 miles per hour (mph) over dry, level, hard-surfaced roads.

3.4.7 Acceleration. Vehicle shall have a minimum average acceleration rate of 2 mph per second for at least 30 seconds. Test shall be in accordance with Federal Motor Vehicle Consumer Information Regulation, 575.106, "Acceleration and Passing Ability" (see 4.4).

3.4.8 Gradeability. Vehicle shall be able to stop and restart three(3) consecutive times during a six (6) minute period while ascending a dry smooth concrete slope having a minimum grade of 30% without stalling or over-heating. Test conditions to be in accordance with Federal Motor Vehicle Consumer Information Regulation, 575.106 (d) conditions and procedures, (1) vehicle, road and ambient conditions, (see 4.4).

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3.4.9 Fuel range. The ambulance shall be capable of being operated for at least 150 traveled miles without refueling when operating, under encountered environmental conditions. (see 3.6.4.4 and 4.4.3).

3.4.10 Fording. The ambulance shall be capable of several two minute fordings test up to 12 inches depth of water while keeping the patient compartment dry. The vehicle shall permit fording without danger of stalling.

3.4.11 Vehicle physical dimensional requirements.

3.4.11.1 Length. Overall length of the ambulance shall not exceed 22 feet including bumpers, but excluding rear step.

3.4.11.2 Width. Overall width of the ambulance shall not exceed 96 inches, for types I and III, and 80 inches for type II, nor be less than the overall width of the rear tires.

3.4.11.3 Height. Overall height of the ambulance at curb weight (3.5.1) shall not exceed 110 inches, including roof mounted equipment, but excluding two-way radio antenna.

3.4.11.4 Ground clearance. The lowest part of the vehicle when loaded to the G.V.W.R. shall have minimum of 6 inches ground clearance. A higher clearance may be specified when necessary for off-highway operations.

3.4.11.5 Angle of approach, ramp breakover, and departure. The vehicle with bumpers and step in place shall have an angle of approach of not less than 16°, angle of ramp breakover of not less than 10°, and an angle of departure of not less than 10° when loaded to G.V.W.R. and measured in accordance with SAE J689a.

3.4.11.6 Turning diameter. Righthand and lefthand turning diameters of the vehicle shall not exceed 2.9 feet per foot of vehicle length in accordance with SAE J695a. Both turning diameters shall be measured at the body corners.

3.5 Vehicle weight ratings and payload.

3.5.1 Curb weight. Curb weight shall include the weight of the complete ambulance; chassis, cab, and body; including all permanently attached or mounted devices, equipment, oxygen cylinders, basic accessories; full complement of fuel, lubricants, and coolant.

3.5.2 Payload allowance. A minimum of 1,000 pounds payload allowance shall be provided over and above the curb weight of the vehicle as specified. The payload shall consist of personnel, patients (computed at 150 pounds per occupant), and miscellaneous support equipment and devices appropriately distributed within the vehicle as designed.

3.5.3 Gross Vehicle Weight Rating (G.V.W.R.) The ambulance gross vehicle weight rating shall be in excess of the combination of weights of the vehicles curb weight and payload weight. Manufacturers shall provide a rating label showing the actual gross vehicle weight rating (G.V.W.R.) of the vehicle.

3.5.4 Weight distribution. The curb weight (3.5.1) of the ambulance shall be equally distributed ± 5 percent over the right and left tire of the same axle when on a level surface. The weight distribution of the fully loaded ambulance on level surface shall be such that not less than 30 percent of the vehicles weight is on the front suspension, (see 4.3.1).

3.5.5 Ratings. Vehicle and component ratings shall be the manufacturer's published ratings, and shall not be raised to meet the requirements of this specification. When ratings are not published, the verification of the equipment manufacturer's rating shall be made available to the appropriate offices of the procuring activity.

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3.5.6 Cab to axle (CA), type I vehicle. Cab to axle (CA) dimension for type I vehicle chassis shall permit a minimum of 58 percent of the outside body length forward of the rear axle centerline, plus 1 to 3 inches for cab to body clearance. Bodies designed with wheel openings shall have the rear wheels centered, ± 2 inches within the opening.

3.6 Chassis, power unit, and components.

3.6.1 Chassis-frame. The chassis-frame and components shall be sturdily constructed and shall be capable of withstanding the strains of on-off road service, any special service, and equipment requirements specified.

3.6.2 Vehicle lubrication. The chassis components, devices, accessories, and equipment requiring lubrication maintenance shall be fully equipped with lubrication fittings.

3.6.3 Power unit, engines.

3.6.3.1 Power unit. The power unit shall meet or exceed the required vehicle performance specified herein (see 3.4), at not more than the engine manufacturer's recommended operating engine speed. It shall be of such design and construction that it will give an even flow of power at all r.p.m. without undue vibration, strain, or overheating of engine parts.

3.6.3.2 Engine low temperature starting. Unless otherwise specified, engine shall start satisfactorily without aid of preheating devices at the minimum of -20 F. Auxiliary heating and/or starting devices shall be provided to meet starting requirements at the option of ordering activity.

3.6.3.2.1 Power plant heaters, min. -30°F. startability. When specified in 3.15.3-16 (see 6.2), manufacturer's standard external or in-block power plant heater(s) shall be furnished for the engine coolant, oil pan or battery when necessary. The vehicle so supplied and furnished shall be capable of starting and operating normally, from -30°F. after standing 18 hours without benefit of solar heat.

3.6.3.3 Gasoline engine. Engine shall be a liquid-cooled eight cylinder internal combustion engine with a compression ratio, which permits efficient operation on either low lead (0.5 gram of lead per gallon, maximum) or unleaded regular gasoline. Engine horsepower and torque requirements shall be sufficient to comply with requirements of 3.4. Net horsepower ratings shall conform to SAE J245.

3.6.4 Power unit components.

3.6.4.1 Oil filter. The oil filter shall be the manufacturer's standard for the engine offered.

3.6.4.2 Air filter. The air filter shall be the manufacturer's standard for the engine offered.

3.6.4.3 Emission controls. Vehicles and engines shall comply with the regulations of the Environmental Protection Agency governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect the year of manufacture. California and/or high altitude (3.15.3-17), pollution control requirements shall be provided when specified (see 6.2), California controls and test shall be provided those vehicles destined for operation in the State of California.

3.6.4.4 Fuel system. The fuel system shall conform to MCSR, Subpart E, 393.65 and 3.4.9 herein. Fuel tank capacity shall be not less than 20 gallons. If more than one tank is furnished, tanks shall be interconnected, and shall have a dash-mounted fuel gauge controlled by a selective switch, which will permit separate fuel level readings for each tank. Class 2 ambulance shall have the fuel tank(s) protected by a metal shield (skid-plate) and shall be located in an area which will be free from hazardous encounters in off-road, cross-country operation.

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3.6.4.5 Cooling system. The cooling system shall be liquid pressurized, forced-circulation type. It shall consist of the manufacturer's heaviest-duty components of sufficient design and capacity to maintain the engine at optimum safe temperature, under all operating conditions specified, without any loss of coolant or overheating of engine and components. Optimum engine temperature shall be maintained with the vehicle loaded to the G.V.W.R. and continually operated at all driveable altitudes and grades in ambient temperatures ranging from -20° to 125°F.

The cooling system shall permit idling of the engine (including high idle at full air conditioning load) under an ambient of 95° F., for a period of 30 minutes without overheating. The cooling medium shall be thermostatically controlled at a minimum of 180 ° F. and suitable for operation with permanent type of anti-freeze solutions. Easily accessible drain outlets shall be provided, to allow complete cooling system drainage.

3.6.4.5.1 Compensating and coolant recovery system. The vehicle shall be equipped with a closed, air-free, liquid-state coolant level indicating overflow recovery and compensating system. The system shall have a hermetical sealing pressure radiator cap, with a spring-loaded vent valve; a minimum 3 pints capacity, vented, unbreakable, translucent reservoir with connecting hose.

3.6.4.5.2 Anti-freeze. All liquid cooling systems shall be protected with permanent-type anti-freeze to a minimum of -20°F., and shall be sufficient to prevent freezing at the lowest ambient temperature enroute to and at the vehicle destination. The cooling system shall be labeled to indicate degree of protection provided.

3.6.4.6 Exhaust system. The exhaust system shall conform to the Motor Carrier Safety Regulations that pertain to buses. The system shall be designed to permit the engine to be idled while vehicle is standing with minimal exhaust fumes and contaminants entering the body interior. Each exhaust system shall be suspended using not less than 3 hangers, and conform to the sound level limits as specified in 3.4.3.

3.6.5 Drive train.

3.6.5.1 Drive train components. The drive train and components torque capacity shall meet or exceed the maximum torque developed in lowest gear ratio by the engine. Transmission and controls shall comply with FMVSS 101 and 102.

3.6.5.2 Automatic transmission. Unless otherwise specified (see 3.6.5.2.1), a continuous drive, automatic transmission shall be provided. Transmission shall provide not less than three-forward and one-reverse speed, and shall be equipped with the related heavy-duty components such as oil-cooler, control devices, and springs when option on model is offered.

3.6.5.2.1 Manual transmission. (type I class 2 only) When specified (see 6.2) a four-forward speed and one reverse speed manually shifted, synchronous type transmission with power-take-off (P.T.O.) opening shall be provided. A permanent shift pattern located on the shift knob or decal within the cab shall be furnished.

3.6.5.3 Transfer case for class 2 vehicle. A two speed transfer case shall be provided 4 x 4 class 2 vehicle. A permanent shift pattern located on the shift knob or decal within the cab shall also be furnished. Unless otherwise specified (see6.2), full-time four-wheel-drive system or continuous drive transfer case maybe furnished. Front wheel drive hubs (semi-automatic, see 3.6.5.9) and special traction (rear-end) differential (3.6.5.8) shall be furnished as specified.

3.6.5.4 Clutch. Vehicles with manual transmissions shall be furnished with the manufacturer's largest heavy-duty clutch available for the engine and model offered.

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3.6.5.5 Drive-line. The drive-line (shaft; U-joints, etc.) shall be balanced and supported to perform throughout the design speed range without whipping or vibrating. Modification resulting from lengthening the wheelbase shall be of chassis manufacturer's approved design.

3.6.5.6 Axle, ratings, ratios. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when vehicle is loaded as specified. When specified (see 6.2), that the vehicle is used in mountainous terrain, manufacturer's shall provide their accomodating higher axle ratio or equivalent engine-transmission, or axle combination.

3.6.5.7 Brake systems, service, and parking. Service brakes shall be power assisted, front disc-drum rear, split or dual hydraulically actuated brakes. Brake systems and components furnished shall comply with 3.4 requirements.

3.6.5.8 Special traction (rear end) differential. A positive traction, limited slip, or automatic locking type differential shall be furnished, except class 2 vehicle when a full time or continuous 4-wheel drive device is furnished.

3.6.5.9 Drive hubs, front wheel, (4 x 4, class 2). Unless continuous drive as in 3.6.5.3, is furnished, semi-automatic front wheel locking hubs shall be provided.

3.6.5.10 Suspension. Vehicle shall be equipped with a matched set of springs, torsion, or air suspension system. Components shall have a rated capacity in excess of the load imposed on each member, Class 1 vehicles springs shall be chassis manufacturer's lowest clamped deflection rating for the model offered (for better riding quality). Suspension stabilizer devices shall be furnished when available and applicable to the model chassis furnished.

3.6.5.11 Spring stops. The manufacturer's standard spring bumper's and axle stops shall be furnished. The front stops and bumpers shall prevent the front axle from striking the engine or oil pan under all conditions of operation.

3.6.5.12 Shock absorbers. Shock absorbers, double-acting type, heaviest-duty available for model offered shall be furnished on the front and rear axle of all vehicles.

3.6.6 Steering. Steering system shall be the manufacturer's recommended power assisted design with a separate oil cooler if required for adequate cooling under all operating conditions specified. Steering system shall achieve the turning diameter specified in 3.4.11.6.

3.6.7 Wheels. Unless otherwise specified (see 6.2) types I and III vehicles may be equipped with dual rear wheels. Type II shall be equipped with single wheels front and rear. Wheels shall conform to the recommendations of the Tire and Rim Association Inc., and shall be alike in type, size and load rating for all wheels on the vehicle including the spare.

3.6.8 Tires. Unless otherwise specified (see 6.2), tires shall be regular highway tread, of the largest size that can be placed on the vehicle without interference with the wheel housings, steering, and suspension systems (see 3.6.11). Tires furnished shall not be less than D load range requiring the minimum cold inflation air pressure to support the weight on each tire measured at the ground.

3.6.9 Inner Tubes. When tube type tires are furnished, inner tubes shall conform to ZZ-I-550 and be in accordance with the recommendations of the Tire and Rim Association, Inc.

3.6.10 Spare tire, and storage. One inflated spare wheel/tire assembly shall be furnished. The spare shall be stored in a weather protected area that provides access without removal of patients from the compartment, or outside the ambulance easily available for quick replacement when needed. If in a exterior compartment, the closure panel shall be fitted with a suitable latch or lock; loose or removable panels shall not be acceptable. When in place, the spare assembly and tools shall have a quick action restraining device to eliminate noisy movement.

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3.6.11 Tire chain clearance. Tire chain clearance for all driving wheels shall be provided in accordance with SAE J683a.

3.6.12 Wheel-tire balancing. Wheel/tire, hubs, and brake drum assemblies of the vehicle shall be in balance to the speed specified in 3.4.6.

3.6.13 Tools (tire changing). Vehicle shall be furnished with tools required for changing spare mounted tire/assembly, at any wheel with the tire flat. The tools shall be the minimum of a jack, jack handle and wheel nut wrench. The jack, without blocking, shall be capable of raising any wheel of the loaded ambulance to a height adequate to permit removal and replacement of the wheel/tire assembly. Tool storage shall be provided.

3.6.14 Hub caps. Manufacturer's standard hub cap or wheel cover shall be fitted on each grounded wheel (except dual wheels) and class 2 vehicles.

3.7 Electrical systems, and components.

3.7.1 Electrical systems. Ambulance shall be equipped with, but not limited to complete and operative 12-volt potential electric generating, starting, and lighting systems, specified electronic equipment and devices; including 110 volt A.C. power supply, and applicable accessory wiring systems. The electrical systems and equipment shall conform with FMVSS, applicable requirements of Motor Carrier Safety Regulations, Part 393 Subpart B and the recommendation of SAE, while complying with the sub-paragraphs herein. The electrical systems shall be reasonably accessible for checking and maintenance. All lamps, electronic devices (housed externally mounted fixtures) shall be chrome-plated or polished metal, rust resistant, and weather proofed. Electrical fixtures attached to the sides of the ambulance below 75-inch height level shall be recessed, tunnel, or near flush mounted not to exceed 2 inches, except such items as spot lamp. Wiring installation shall include RFI traps, to minimize interference to biophysical monitoring equipment.

3.7.2 Wiring. All of the ambulance body electrical equipment shall be served by circuit(s) separate and distinct from the vehicle chassis circuit(s). Body wiring shall be thermoplastic harness type, color coded or numbered, grease, oil, and moisture resistant, routed in protected locations, neatly and securely fastened, and all apertures properly grommetted for passing wiring. Solderless insulated connectors shall be provided. The body electrical wiring shall incorporate overload protective devices of the automatic reset, circuit breaker type. In addition, one(1) space for single pole, 15-ampere circuit breaker shall be furnished for future use. The circuit breakers shall be readily available for inspection and service. Complete wiring diagrams (schematics) with identification codes for the standard and the optional equipment furnished shall be supplied with each ambulance, and may be included with manuals (3.20). Minimum size of the schematics shall be 8 1/2 inches by 11 inches.

3.7.3 Ignition system. Spark ignition engines shall be furnished with the manufacturer's current standard ignition system. To permit fording, the ignition system shall be moisture proof (3.4.10). The ignition switch shall comply with FMVSS 101, and 102, and be supplied with two keys.

3.7.4 Windshield wipers and washer. Vehicle shall be equipped with dual, electric, multispeed (two or more), windshield wipers and washers complying with FMVSS 104.

3.7.5 Horn. Chassis manufacturer's standard electric horn shall be furnished.

3.7.6 Electric generating system (alternator). The generating system shall be capable of supplying the maximum built-in D.C. electrical current requirements of the ambulance, but not less than a 105 ampere rated capacity alternator.

3.7.7 Battery. Dual 12-volt potential battery system with a switch allowing the electrical system to function from either one or both batteries simultaneously or dual battery with isolator device allowing simultaneous battery charging shall be furnished. The batteries provided shall have a minimum performance rating of 350 amps @ 0° cold crank with at least 100 minutes reserve capacity each. Battery ratings shall comply to table 1, of W-B-00131J. Batteries shall be located in a ventilated area sealed off from occupant compartments, easily accessible for checking and removal.

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3.7.8 110 Volt AC electric power. An AC power inverter to provide the following:

a. Continuous 500 watts, single phase, constant 60 cycle 110 volt power output at all engine speeds to operate bio-medical electronics and patient care equipment to be furnished, and;

b. When specified (see 3.15.3) a minimum continous 3000 watts single phase, constant 60 cycle 110 volt power output to also operate rescue tools, lights and other emergency equipment shall be provided, and;

c. On-off switch with lighted indicator located at a 110 volt receptacle shall be provided.

3.7.9 Electrical 110 volt receptacles. The patient compartment shall be furnished with 3-wire grounded duplex receptacles. Receptacles shall be located at the lower torso area of the primary and secondary patients, for the use of electric powered medical devices.

3.7.10 Radio-interference suppression. Ambulance shall be radio-interference suppressed in accordance with chassis manufacturer's standard practice and SAE J551a and J552a. Specified or furnished equipment and devices emitting radio interference shall be suppressed to the same level as the vehicle chassis.

3.8 Lighting, ambulance exterior and interior.

3.8.1 Ambulance exterior lighting. The basic exterior vehicle lighting shall include: headlights; parking lights; directional signal lights; tail and stoplight; sidemarker lights; license plate lamp; back-up lights; hazard warning lights; clearance lamps, when applicable; emergency beacon light and corner warning lights 3.8.2; flood lights 3.8.3, and spot light 3.8.4.

3.8.2 Warning lights. Unless otherwise specified (see 6.2), a roofmounted 360° rotating dome beacon lamp containing four sealed beam bulbs, each having a minimum 35,000 candle power, with alternating lens colors, two white and two blue or strobe light capable of producing a minimum of 100,000 candle power, shall be provided and located near the center of vehicle or body. The upper body or roof corners shall also be provide with strobe or alternate flashing pattern blue color lights. Corner warning lights dome, or tunnel, (two on the front facing front and two on the rear facing rear) shall be seal beam type having a minimum of 300 candle power output each. In addition at least two blue flashing warning lights, shall be located in, on or behind the vehicles front grill. Warning lights shall comply with SAE J845 and J595b. Warning light mountings shall be provided with reinforcements at point of attachment to vehicle. Warning lamps shall not be obstructed by opened doors.

3.8.3 Flood and loading lights (exterior). "Sealed flood lightings units shall be located at the rear, left and right sides of the ambulance. Flood lights shall be integrally mounted below the roof line, but not less than 75 inches above the ground and unobstructed by open doors. Flood light(s) shall have a minimum of 800 candle power output per side, and conform with SAE J598a. Flood lighting at the rear shall be arranged to illuminate the ground area immediately surrounding the loading doors." The switches shall be shielded and control each side separately. (3.8.7.1)

3.8.4 Spotlight. A single 360° swivel, clear spot lamp of not less then 75,000 candle power, mounted on the cab roof, over the windshield area (between but clear of siren/speakers) or two piller or cowl types may be substituted, convenient for the driver and assistants use.

3.8.5 Ambulance interior lighting. The basic interior ambulance lighting shall include: driver compartment dome light; instrument panel lights and indicators; master switch panel or console lights; other warning lights; "Door-Open indicator; and glove box light, when applicable. The patient compartment shall include: overhead or dome lighting 3.8.6; type I and III vehicle, step-well light(s), and switch panel lighting; 3.8.7. Lighting shall be designed and located so that no glare is reflected from surrounding areas to the driver's eyes or his line of vision, from instrument and switch control panels or other areas that are illuminated while the vehicle is in motion.

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3.8.6 Patient compartment illumination. Illumination shall be adequate throughout the compartment, and shall provide a minimum intensity of 40 foot candles at the floor level of the compartment. At least four recessed and near flush dome lights shall be furnished. Dome lights shall be located one each side of the center line of the compartment ahead of the primary patient and at the foot. A reduced lighting level shall be provided by a 3-way switch to control compartment lighting or by a second system of low intensity lights. Dome and step-well lights shall be automatically activated when opening the entrance doors in addition to the front switch panel and drivers compartment switch panel (3.8.7.1).

3.8.7 Switches and control devices. Instrument panel, switches and control devices shall be located for left hand drive vehicle. All switches and control devices shall be clearly visible for the ambulance personnel; identified plainly, and permanently marked; grouped according to function; mounted in lighted panels, or consoles and installed in a manner to facilitate removal and serving. All controls shall comply with FMVSS 101a.

3.8.7.1 Driver compartment switches. The following switches and controls shall be located within the normal reach of the driver without excessive movement of his head or eyes from his driving field-of-view. These shall include and control:

All exterior lights
Communication(2-way radio, intercom, public-address)
Environmental equipment, (cab and patient compartment master switch).
Siren (master control, foot control or push-button)
Driving instruments and vehicle operating controls
Interior lights (switch panel, instruments, etc.)
Emergency master switch
Throttle (manual hand operated, if required for AC power, or specified)

3.8.7.2 Patients compartment switches. The following switches and controls shall be furnished when applicable, and be easily accessible to the technician when seated at the patient's head. The forward panel shall control:

Interior lights (on-off, panel light, etc.)
Heating
Ventilation
Air Conditioning
Communication (2-way radio microphone, intercom)
A.C. 110 volt power supplies (on-off switch)

Switches shall be located at the rear doors, and be operable from ground level reach to control the following:

Exterior area flood lights (separate switches)

3.9 Cab-body driver compartment and equipment.

3.9.1 Driver's compartment, cab-body structure. All cab compartments shall be of sufficient size to accommodate two 5 percentile to 95 percentile males (driver and assistant) with space to perform driving and control activities. The cab (type I) or the cab-body (types II and III) shall be organized and designed with the specified and required equipment and accessories for ease of operation and safety. The cab and cab-body shall comply with Federal Motor Vehicle Safety Standards 101, 102, 103, 104, 107, 108, 111, 113, 205, 206, 207, 208, 209, 210, 211, and 302. Types II and III vehicles of the integral cab-body design shall essentially be equipped with front doors and window openings equal to the type I conventional truck cab, and conform to 3.9.2 thru 3.9.4. All ambulance types shall be provided with a bulkhead or partition between the cab or driver's compartment and the patient's compartment (see 3.10.2 and 3.10.15).

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3.9.2 Cab-body provisions. Cab-body section shall provide a right and left side weatherproofed forward hinged door, with crank operated side windows and crank or push open vent windows; door stops; external key operated door locks, with two sets of keys; standard cab compartment insulation and sound deadening; trim panels or closed panels, and headliner; floor covering; panel mounted instruments; seat(s); painted interior exposed surfaces; hardware and other exterior exposed metal trim chrome plated, stainless steel or anodized aluminum; and furnished with at least the following equipment:

(a) Dual sunvisors (padded).
(b) Armrests, mounted on each side door if available.
(c) Compartment ventilation, other than windows.
(d) Key-operated ignition switch.
(e) Ammeter.
(f) Fuel gage(s).
(g) Oil pressure gage.
(h) Engine temperature gage.
(i) Speedometer with odometer.
(j) Environmental controls (heater-defroster/air conditioner, etc.).
(k) Seat belts; and driver's shoulder harness.
(l) Dual outside mirrors (3.9.5).
(m) Cab lighting and controls.
(n) Tinted windshield.

3.9.3 Cab compartment driver and assistant seat. The driver's compartment shall be furnished with two individual bucket-type seats (driver and assistant). The seats shall be frame constructed with cushioned springs or foam rubber, padded, and upholstered to provide maximum riding comfort. The seats shall be covered with fire-retardant, washable, artificial leather or plastic; nonabsorbent material. Driver's seat shall be adjustable a minimum of four inches forward or backward.

3.9.4 Controls and operating mechanism. All controls and operating mechanisms shall be located for left-hand drive. Lever controls and equipment, items and devices shall be installed, located, and stowed for the convenience of the purpose intended, shall not interfere with the ambulance personnel or patients ingress or egress of respective compartments.

3.9.5 Outside rear view mirrors. Dual below eye level type outside rear view mirrors with a minimum of three supporting arms, and fully adjustable mirror head, each with not less than 70 square inches of reflective glass area shall be furnished. Mirrors shall include a minimum 4-inch diameter stick-on convex curved wide vision mirror glass mounted on the lower portion of each mirror. Hardware and mirror heads shall be of polished metal.

3.9.6 Bumpers and steps. A sturdy front and rear, full-width vehicular or body bumper(s) secured to the vehicle's chassis-frame shall be provided. Step shall be provided at back door opening of the patient compartment if the floor is more than 18 inches above the ground. If step is required, the height from the ground to the first step shall be equal distance $\pm$ 2 inches from the ground to the floor of the patient compartment. The step shall be designed to prevent the accumulation of mud, ice, or snow, and have a non-skid surface with a minimum depth of 5 inches (tread). Step may be the open grating type, provided it is not located or exposed to the interior of the ambulance when the door(s) is closed. All necessary steps shall be at least the width of the door opening for which they are provided.

3.9.7 Fenders. Fenders or inner weather shields shall be provided over all wheels and tires.

3.9.8 Engine hood. Engine hood and cowl shall be fitted to prevent leakage from precipitation, heat, smells, and noise from entering the interior of the cab and body. Cab compartment hood, types II and III ambulance shall open sufficiently for easy access to engine components requiring routine maintenance.

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3.9.9 Cab connecting bellows, for type I vehicle. A flexible, weathertight bellows of rubberized nylon construction shall be provided between the cab body and the containerized modular body. Window in the cab body shall be of the sliding type within attached bellows aligning and connecting with the modular body window opening, and shall generally conform to requirements of the partition (see 3.10.15).

3.10 Ambulance body or patient area.

3.10.1 Body accommodations. The ambulance body proper or patient compartment or area shall be sufficient in size to transport occupants as specified in plan A or B (see 3.1.5), and accommodate and store all the stretchers, cots, and litters through the range of dimensions as specified in table I. There shall be space around the patients to permit a technician to administer life supporting treatment to at least one patient during transit (see dimensional parameters 3.10.4).

Table I. Ambulance Stretchers, Cots, and Litters

Stretchers, Cots, and Litters	Dimensions (In Inches) Length Minimum	Width Minimum	Bed Height Maximum
Style 1 -- Wheeled Cot (Elevating)	75	22 1/2	15 [1]
Style 2 -- Wheeled Cot-Bench (Non-Elevating)	75	21 1/2	19 [1]
Style 3 -- Folding Stretcher or Combination Stretcher Chair	73 1/2	19	8 1/4
Style 4 -- Navy Stokes Litter (Resting on Floor)	84 3/4	23 1/2	7 1/2
Style 5 -- Standard Army and NATO Litters (With Poles) [2]	90(-0,1/4)	23(-6/10,-3/4)	6 3/4 (-0)

Notes: [1] Measured to top of positioned 3 inch thick mattress.
[2] Dimensions of Army and NATO Litters are in accordance with North Atlantic Treaty Organization Standardization agreement STANAG No. 2040.

3.10.2 Partition with access between compartments (types II and III). When specified in 3.15.3, ambulance shall be provided with access to the patient's compartment without leaving the vehicle. When so equipped, the door (sliding or hinged) between the two compartments shall be secured on the driver's side of the compartment. Partition in general shall otherwise conform to 3.10.15.

3.10.3 Technician seating. The technician shall be provided with space and a seat equipped with safety belt and headrest, not less than 18 inches deep by 18 inches wide and 12 to 18 inches high (par. 3.11.1.1). Technician shall be seated at the head of the primary patient at the bulkhead or partition behind cab compartment, (see 3.1.5 and 3.11.4). Space under the seat may be designed as storage compartment, unless utilized for other equipment such as heater or vents, etc.

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3.10.4 Patient compartment interior dimensional parameters. The patient's compartment shall provide, but is not limited to, a minimum of 300 cubic feet of space, less 10 percent deviation for cabinets, while complying with the following:

Length; Measured from the partition to the inside edge of the rear loading doors at the floor, shall be at least 116 inches in length. This length in the compartment shall provide at least 25 inches and not more than 30 inches of unobstructed space at the head of the primary patient, measured from the technician's seat back-rest to the forward edge of the style 1 cot.

Width; The width of the compartment after cabinet and cot installation, shall provide at least 18 inches in type I and III, and 12 inches in type II ambulance of clear walkway between cot(s) and the squad bench, and at least 25 inches (width) of kneeling space (for the technician) along side the primary cot, measured at the floor for a height of 9 inches, from the forward leading edge (corner) half the length back of the (primary) wheeled elevating cot.

Height; Minimum 60 inches. The patient compartment shall provide at least 60 inches height over the patient area, measured from floor to ceiling, exclusive of cabinets or equipment, and symmetrical corners and edges.

3.10.5 Body, general construction. The body, type I modular unit, and the type II and III integral cab-body shall be of prime commercial quality metal or other material with strength at least equivalent to all-steel. Wood shall not be used for structural framing. The exterior of the body shall be finished smooth with symmetrically rounded corners and edges except for rub rails, presenting a modern and aerodynamic appearance, and shall embody provisions for doors and windows specified herein. Ambulance body (prior to outside attached devices) as a unit shall be designed and built to provide impact and penetration resistance, and shall be of sufficient strength to support the entire weight of the fully loaded vehicle on its top or side if overturned, without crushing, separation of joints, or permanently deforming roof bow or reinforcements, body post, doors, strainers, stringers, floor, inner linings, outer panels, rub rails, and other reinforcements. As evidence that ambulance body meets the above criteria, the body manufacturers, (fabricated, modified, or conversions), excluding the conventional cab shall furnish for each body model (type) a certification that the ambulance body meets Static Load Test Code for School Bus Body Structure, Standard No. 002.

3.10.6 Ambulance body structure. All parts of the ambulance body, where applicable shall be fastened together with rust-resistant fasteners in a manner which will preclude loosening of bolts, screws, and rivets, and cracking of welded joints. Metal tapping plates welded to the walls or ceiling structures to provide firm securing for installed equipment (cabinets, benches, partition, cylinders) shall be employed in the ambulance structure. Self-tapping screws shall not be used where subjected to stress. The body, roof, and panel joints shall be watertight. All openings between chassis and occupant carrying compartments due to alternation or construction shall be sealed, including bulkhead space between cab and body, type I (see 3.9.9).

3.10.7 Body mounting. Heavy rubber or antisqueak material shall be utilized between body frame and attaching chassis. Reinforcements or filler blocks shall be used where mounting devices might otherwise deform frame flanges. Mounting devices shall be locked units which will minimize loosening, but which may be tightened if necessary. Types I and III ambulance body shall have at least 4 high-strength 5/8-inch bolts to attach the body brackets to the frame chassis on each side, mounted so as to prevent any side movement of the body. The ambulance body of type I vehicle shall not be welded to the frame at any point, and shall be interchangeable with other manufacturers vehicle chassis having the same cab to axle (CA) dimension.

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3.10.8 Doors. Two door openings, other than the doors for the driver cab compartment shall be provided. There shall be a door(s) opening on the right forward side and a two door loading opening, at the rear of the patient compartment. A forward hinged single door, double doors or a sliding type door shall provide a minimum right side-door opening of 30 inches wide, and 54 inches high for types I and III and 42 inches high for type II. Rear loading doors shall cover a clear opening of not less than 46 inches in height and 50 inches in width for types I and III, and the manufacturer's standard for type II. Both doors shall be hinged at the outer edges and open back towards the side of the vehicle. The ambulance body doors shall be equipped with not less than 250 square inches of safety glass area per door. Doors shall be designed for easy release, but prevent accidental opening with an inter-locking system that functions even when doors are not completely closed. A "Door-Open" warning device shall signal (indicator or buzzer) when doors are not closed. Each door shall have effective compression or overlapping seals to exclude dust, water, and air. Doors may contain and be equipped with recessed compartments as applicable to the interior for storage of supplies and devices.

3.10.9 Door latches, hinges, and hardware. Door latches, hinges, and hardware shall comply with FMVSS 206. When doors are open, the hinges, latches, and door-checks shall not protrude into the access area. All doors shall have hardware or devices to prevent inadvertent closing; a grab handle on the inside of each door, in addition to a door operating handle; door stops to prevent damage to body sides; a handle with latches operable from inside and outside of the body with one external operated lock with key per door opening. Hardware and other exposed interior trim shall be chrome plated, stainless steel, or anodized aluminum. All hangers or supports for equipment or other items shall be mounted as flush as possible with the surrounding surface when not in use.

3.10.10 Floor. Floor shall be at the lowest level permitted by clearances, but not more than 33 inches from the ground; exception of class 2 vehicles. It shall be flat and unencumbered in the access and work area. Floor shall withstand a distributed load of 150 pounds per square foot. Metal floor shall be reinforced to eliminate "oil canning" and insulated against outside heat and cold. The floor of the patient's compartment shall be either (1) of sheet metal; minimum 20 gage, and reinforced with 1/2-inch thick five-ply water-resistant plywood, or (2) marine grade plywood not less than 3/4-inch thick, both supported by body framework. There shall be a minimum of voids or pockets at the floor to side wall areas, where water or moisture can become trapped to cause rusting and unsanitary conditions. Voids and pockets shall be filled with sealer or caulking compound. Flooring shall extend the full length and width of the compartment or body (including space under the cabinets).

3.10.11 Floor coverings and color. Floor covering shall be provided that harmonizes with the interior of the patient's compartment. The floor covering shall be one piece, no-wax, skid resistant linoleum or vinyl not less than 1/16-inch in thickness; permanently applied, and shall extend the full length and width of the compartment. Covering joints at side walls, where side panels and covering meet shall be sealed and bordered with rustproof, corrosion-resistant metal cove moulding.

3.10.12 Stepwell (side door). Steps shall be provided at door opening if the floor is more than 18 inches above the ground. Stepwell, when applicable shall be the enclosed two-step type. Height of bottom step except class 2 vehicles shall not exceed 18 inches in types I and III vehicles and 22 inches in type II. Stepwell of types I and III body shall be lighted (see 3.10.5 interior lights), and all step surfaces shall be covered or taped with anti-slip material.

3.10.13 Wheelhousings. Wheelhousings of types I and III vehicle shall include splash shields under the body. The splash shields shall extend around the top of the wheel and to the bottom of the body and side skirting. Wheelhouse opening shall allow for easy tire removal and service. Chassis manufacturer's standard wheelhousing will be acceptable for type II.

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3.10.14 Windows. Unless otherwise specified (in 3.15.3), the patient's compartment shall not have windows except the viewing panel in the partition or bulkhead(s) (3.10.15) and in the doors (3.10.8). Glazing shall comply to FMVSS 205.

3.10.15 Partition. A load bearing bulkhead or partition in all ambulance types shall be placed between the driver and patient's compartment. Types II and III vehicle shall have the partition located in back of the driver's seat rear-most position. The partition shall incorporate a shatterproof viewing window of at least 100 square inches for visual check of conditions in the patient's compartment by the driver. Viewing window shall be treated or located in the partition or bulkhead to prevent the lighted domelights in the patient's compartment from interferring with the driver's night vision and visability. There shall be a means of voice communication between the driver and the technician by way of opening the partition viewing window(s), and by an intercom system (3.14.5). Partition and storage compartment shall be constructed and secured at sides, ceiling, and floor by metal channels or angles that are bolted or welded to the tapping plates, located behind side walls and ceiling panels. Partition shall be firmly secured to withstand shock and have roll-bar characteristics. Partition with access door, see 3.10.2.

3.10.16 Insulation. The entire body; sides, ends, roof, floor, and patient's compartment doors shall be sufficiently insulated to enhance the environmental criteria specified (3.4.2 and 3.13), and minimize conduction of heat, cold, or external noise entering the vehicle interior. The insulation shall be vermin and mildew-proof, fire-retardant, non-hygroscopic, and a non-settling type. Undercoated floors will be considered to have sufficient insulation for the floor area.

3.10.17 Interior surfaces. Interior of the body and the exterior of the storage cabinets shall be free of all unnecessary projections. Exposed edges and corners shall be broken with a radius or chamfer, beaded, hemmed, or flanged, or otherwise treated to minimize sharp edges and corners. The finish of the entire patient's compartment, including stowage cabinets and equipment shall be impervious to soap and water, disinfectants, mildew, and shall be fire-resistant. Interior body lining and cabinetry, excluding the cab compartment(see 3.9), shall be of high strength metal panels,reinforced glass fiber, vinyl coated wood, plywood or aluminum, plastic panels, plastic impregnated panels, or polyester sheets of equivalent strength. Ceiling headlining may be of vinyl upholstery material.

3.11 Patient compartment facilitations.

3.11.1 Interior stowage accommodations. The interior of the patient compartment shall provide, but shall not be limited to, a minimum of 30 cubic feet of enclosed stowage cabinetry, compartment space, and shelf space which shall be conveniently located for medical supplies, devices, and installed systems as applicable for the service intended (reference the U.S. Department of Transportations, (DOT) report; Ambulance Design Criteria.) Enclosed compartments and spaces shall be located at, in, or on the partition, side walls, overhead, squad bench, technician seat and doors. Compartment(s) under floor with opening panel(s) inside patient compartment shall not be acceptable.

Whether specified herein or not, stowage shall be required for the following items:

Item	Approximate Area in Cubic Feet
Medicine dispensary cabinet(s)	5
Medical supplies cabinet	8
Linen supplies	4
Oxygen installation (main) (see 3.12)	6
Oxygen unit (portable)	1
Telemetry equipment	2
Radio equipment & Antenna	1
Storage, Misc.	2
Vacuum aspirator unit	1
Air conditioning unit	as required
Heating system	as required
Backboards (long and short)	as required
Stretchers storage (folding and scoop)	as required
Switch panels & controls	as required

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3.11.1.1 Location of equipment and supplies. Shall be dictated by the relative importance of ready accessibility to the technicians, with priority given to items necessary to cope with life-threatening conditions at the scene and in transit. The equipment and supplies necessary for airway care, artificial ventilation, oxygenation, and suction shall be within reach of the technician at the head of the primary litter, and when furnished, those for cardiac resuscitation, control of external hemorrhage, administration of intravenous agents, and the monitoring of blood pressure shall be readily available to the technician at the side of the litter.

3.11.2 Extrication equipment and stowage accommodations. An interior or preferably an exterior compartment shall provide convenient, weather-proof storage space as applicable to the vehicle. Exterior compartment door(s) shall have suitable rust-resistant flush type latch and key lock. Space for the following items shall be provided, and furnished in the extrication stowage compartment when specified in 3.15.3-18:

Item	Approximate area in cubic feet
One wrench, 12" adjustable, open-end	4
One screw driver, 12" regular blade	
One screw driver, 12" Phillips type	
One hacksaw with 12 wire (carbide) blades	
One pliers, 10" vice grip	
One hammer, 5# 15" handle	
One fire axe, butt, 24" handle	
One wrecking bar, 24" combination tool	
One bolt cutter, 39" jaw opening of 1-1/4"	
One portable hydraulic power jack and spreader tool hand-powered, minimum 4-ton capacity	
One shovel, pointed blade (folding type)	
One double-action tin snip, minimum 8"	
Two ropes, manila, 50' x 3/4" diameter each	
Two pairs gloves (gauntlets) leather	
Two pairs goggles (clear eye protective)	

3.11.3 Stowage compartments and cabinets design. Storage cabinets, drawers and kits shall be easily opened, but shall not come open in transit. For rapid identification of contents, medical supply cabinets above the litter patients shall have shatter-proof transparent sliding door(s). Stowage compartments shall be divided into sections, shelves shall be adjustable, pullout or sliding type, and drawers removable. Cabinet compartment, drawers, and doors (sliding or hinged) shall be automatic latching or closing or fitted with friction holding devices. Oxygen cylinder (3.12.1) shall be within a crashworthy compartment,vented to the outside, mounted with a minimum of 3 restraining devices. Cabinet shelves shall be not less than 10 inches in depth below vehicles belt level. Stowage compartment, cabinets and equipment support area shall be finished in accordance with 3.10.17. Cabinets shall be firmly anchored to tapping plates of the body structure (3.10.6). Tops of the cabinets shall be bordered or surrounded by a lip of not less than 1/2-inch in height, and cabinet shelves of not less than 1/4-inch height.

3.11.4 Squad bench, seats and backrests. The squad bench (required for floor plan A) shall support the style 3 folding/chair stretcher specified (see 3.10.1 and 3.11.5) Squad bench platform shall be not less than 22 inches wide, 72 inches long and 12 to 18 inches high providing at least 18 inches wide padded seat(s), stretcher post cups for legs and wheel cups to hold the specified style 3 stretcher(s), and underneath storage. All seats in the patient compartment shall have padded backrests, including the bench cot when furnished under floor plan B. Padding furnished shall be foam rubber or polyether urethane foam of not less than 3 inches thick, and covered with fire-retardant, washable, non-absorbent artificial leather or vinyl. Backrests shall be padded with foam not less than 1 inch thick, and covered with identical material as used on the seats. Squad bench or cot shall be furnished with 3 safety belts, which may double as cot restraints, (see 3.11.6).

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3.11.5 Stretchers, cots and litters. Ambulance shall be provided with one style 1 cot for the primary patient, and a stored folding chair stretcher (style 3) designed to carry a patient over stairways and other narrow areas (see 3.11.5.1). When floor plan B arrangement is specified (1.2.2) a wheeled style 2 cot conforming to a Ferno Washington Model 409-M Cot-Bench or equivalent shall be furnished secondary patient. Stretchers, cots and litters shall conform to the dimensions of table I (3.10.1) length and width measurements shall be taken at the metal framing excluding joint fittings. Wheeled cots shall include: foot and head pulls; minimum 3-inch thick polyether mattress with vinyl coated nylon fabric (mattress filler and cover material shall conform to MVSS 302) with zippered cover for style 1 and 2 cots. Unless otherwise specified (3.11.5.1 and 3.15.4). Style 3 stretcher, shall conform to a Ferno Washington Model 11, or equivalent with legs, 2-wheels and patient straps.

3.11.5.1 Combination stretcher chair. When specified in 3.15.4, a Ferno Washington Model 107B Combination Stretcher Chair or equivalent shall be furnished in lieu of the stored stairway stretcher specified in 3.11.5. Stretcher shall conform to the dimensions of style 3, of table I (3.10.1). Stretcher shall have fixed posts and wheels, with wheel chair, stair chair and stretcher capabilities, with folding squad handles at foot-end and straps.

3.11.6 Seat safety belts and anchorages. All seats shall comply with FMVSS 207. Safety belts and anchorages shall comply with FMVSS 208, 209, and 210. Seat belts shall have retractor devices in all seat positions in the vehicle, including squad bench.

3.11.7 Litter fasteners. Crash-stable fasteners of the quick release, adjustable type shall secure the specified litters to the floor or side walls. Where a single patient may be centered in the area on the wheeled litter, additional attachments (cups and locks) shall be provided. Attachments shall be near flush with surrounding surface when not in use.

3.11.8 Patient restraint. At least three strap type restraining devices (chest, hip, and knee) shall be provided per stretcher, cot and litter to prevent longitudinal or transverse dislodgement of the patient during transit. Restraining straps shall be not less than 2 inches wide, nylon, easily removable for cleaning, two piece assembly with quick release buckles.

3.11.9 IV holders and hooks for intravenous fluid containers. A rigid telescoping IV pole and holder, detachable type, with a 52 inch minimum height when extended shall be provided the style 1 cot, and one for either the specified style 2 cot-bench or style 3 stretcher. When specified in 3.15.4 two near flush type IV ceiling holders or hooks shall be provided. These shall be located at the side and heads of the primary patient's cot and one at the secondary patient's cot or squad bench.

3.12 Oxygen and suction systems and equipment.

3.12.1 Oxygen; main supply and installation. The ambulance shall have a hospital type piped oxygen system, capable of storing and supplying a minimum of 3,000 liters (see cylinder data 6.8) of medical oxygen. This main oxygen (O_2) supply may be provided by using a single tank ("M" size cylinder). Cylinder(s) shall conform to Federal Specification RR-C-901, DOT specification 3AA2015, and MS-39226-8. The oxygen cylinder(s) shall be located in a exterior vented (using one way check valve type device) storage compartment (see 3.11.3) accessible for replacement preferably from outside the patients compartment working space. Mounting and restraints for oxygen cylinder(s) shall be crash-worthy. The cylinder controls shall be accessible from inside the vehicle. The pressure gauge and regulator shall be visible by way of see-thru-door from the technician's seat, and from the site where the cylinder(s) change is accomplished. The ambulance consignee will install the oxygen cylinder(s) at the time the vehicle is placed in service. Cylinder changing wrench(es) shall be furnished chained within the oxygen cylinder compartment. The contractor shall install all other components and accessories required for the piped oxygen system, which shall include, but not limited to: at the cylinder(s), a reducing valve, pressure regulator having at least a 0-2,000 PSI range (4000 PSI test) and be preset to 50 PSI line pressure, (visible to the technician): non-ferrous piping and low pressure hose suitable for medical oxygen. Oxygen piping shall be concealed, and not exposed to the elements or damage, securely supported, and readily accessible for inspection and replacement. Oxygen shall be piped to valves of two outlets, one duplex oxygen outlet station for the primary patient, and one oxygen outlet for the secondary patient; at least one oxygen wall outlet for the primary patient shall be equipped with a plug-in flow-meter, humidifier, and delivery tube, the second oxygen wall outlet will be used for quick disconnect plug-in devices not requiring humidification. Outlets shall be adequately marked and identified, and not interfere with the suction outlet. The entire system shall be tested (see 4.4.5).

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3.12.2 Oxygen; portable. A portable oxygen (O_2) unit of at least 300 liter ("D" size cylinder, see 6.8) shall have a yoke, pressure gauge, flowmeter (not gravity dependent), delivery tube and oxygen mask(s), and be located near a patient compartment door. The unit shall be capable of delivering an oxygen flow of at least 10 liters per minute. A full spare cylinder of oxygen for this unit shall be furnished and stored. Portable cylinder holder(s) shall be furnished with a quick release securing fittings to allow the use of portable unit outside the vehicle. Oxygen masks (with or without bags) shall be semi-open, valveless, transparent, disposable or easy to clean and decontaminate, in sizes for adults, children, and infants.

3.12.2.1 Artificial ventilation devices, (portable). When specified (code M6) in 3.15.4 two portable, artificial ventilators, squeeze bag valve mask units, complete with face holders for adult, child and infant masks, that operate with either air or oxygen enrichment shall be furnished. Masks shall be transparent, to permit rapid recognition of color change, vomiting, and breathing. Attachment of the reservoir tubes to the intake valves shall permit delivery of 100 percent oxygen. Oxygen powered manually triggered inflation devices are acceptable if they can provide instantaneous flow-rate of one liter per second or more for adults. A safety valve release pressure of approximate 50 cm H_2O for adults and 30 cm H_2O for infants and children shall be provided, and permit the use of 100 percent oxygen.

3.12.3 Suction aspirator system. An engine vacuum operated or electrically powered suction aspirator system shall be provided for the primary patient. The installed suction system shall be powerful enough to provide an air flow of at least 30 liters per minute (1.06 cubic feet per minute) at the end of the delivery tube, and a vacuum of at least 300 millimeters (11.811 inches) of mercury to be reached within 4 seconds after the tube is clamped. The suction line shall be securely fastened, to prevent damaging, but accessible for inspection and repair. It shall be connected to a valve; and a reservoir chamber having a volume of not less than 300 cubic inches, in line between the vacuum source and a single, plug-in, self-sealing valve connection. A plug-in outlet shall be located near the oxygen outlets, but far enough away to permit easy handling of suction and nonbreakable humidifier bottles. Unless otherwise specified, suction system shall include the following: two mating outlet fittings or yokes for the valve connection; one 10-foot medical vacuum hose; suction control valve regulator, vacuum gauge, which shall be connected to a nonbreakable, transparent, suction trap bottle of one liter (1.057 quarts) capacity, and located in the side cabinet (or mounted on rail when furnished) next to the primary patient and technician; one spare nonbreakable suction bottle stored nearby, when disposal bag is not used; one semi-rigid pharyngeal suction tip (nonmetallic), one suction rinsing water bottle; pharyngeal tubes (air ways adult, child, infant); spare sterile suction tip and catheters; one Y connector for tracheal suctioning; and all other devices or accessories specified or required for a ready-to-use aspirator system.

3.12.4 Portable suction aspirator. A portable self contained battery or manual suction unit shall be provided, having at least 12 inches mercury negative pressure, and flow for pharyngeal suction. The portable suction unit shall be fitted with large bore, non-kinking suction tubing and semi-rigid pharyngeal suction tip (nonmetallic).

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3.13 Environmental climatic equipment.

3.13.1 Environmental systems. All ambulances shall be equipped with a complete climate environmental system(s), to supply and maintain clean air conditions and comfortable level of inside temperature in both driver and patient compartments. The various systems for heating, ventilating, and air conditioning may be separate or combination systems, which shall permit independent control of environment within each compartment. The air systems shall be of high volume capacity with low velocity delivery for minimum draft circulation while providing a positive pressure within each closed compartment when the vehicle is in service. Environmental system components shall be readily accessible for service at the installed locations.

3.13.2 Driver's compartment environmental equipment. The driver's compartment shall be furnished with a hot water, fresh-air, high capacity, heaviest-duty (OEM) heater, with dual defrosters, and dehumidifying air conditioner system. Systems shall provide outside air and variable mixtures as desired, circulating conditioned air through the compartment, in compliance with the environmental criteria specified herein.

3.13.3 Patient compartment environmental equipment. The patient compartment shall have heating equipment, ventilation provisions, and air conditioning equipment complying with the environmental criteria specified herein. Unless the reheat-recycle type conditioning system is furnished, the heating system shall be a high-capacity hot water type heater of not less than 25,000 BTU at 150 degrees F. water over air differential, with an air flow of not less than 250 cubic feet per minute.

3.13.4 Heating criteria. When measured midway between the floor and the ceiling the heating system(s) shall have sufficient capacity to heat the driver and patient compartments to a minimum dry bulb temperature of 75°F., within 30 minutes time at 1200 to 1500 RPM engine speed in neutral gear, after a cold soak in an ambient temperature of 0°F. Temperature gradient between the floor and ceiling shall not exceed 10°F., with partition openings closed. Heater(s) furnished shall comply with Motor Carrier Safety Regulations 393.77. Heater air outlets shall be located to provide a comfortably heated compartment.

3.13.5 Air conditioning criteria. When measured midway between the floor and the ceiling, the air conditioning system(s) shall have sufficient capacity to lower the temperature in the driver and patient compartments to 75°F., within 30 minutes time at 1200 to 1500 RPM engine speed in neutral gear, and maintain that temperature while operating the vehicle in an ambient temperature of 95°F. In addition, the cooling capacity of the air conditioning system shall be sufficient to maintain a minimum of 15°F. ambient air temperature differential (outside/inside temperature) under all other operating conditions (see 3.4.2). The main components compressor, condenser(s), evaporator(s) and clutch(s) shall have a balanced system and be in accordance with established practices and standards of International Mobile Air Conditioning Association, Inc. (IMACA).

3.13.6 Ventilation criteria. Ventilation system(s) of the driver and patient compartments shall provide a complete change of air within each compartment every two minutes. Ventilation shall be separately controlled within each compartment. Fresh air intakes shall be located in the most practical contaminant-free air space on the vehicle. Compartments shall be ventilated by the air delivery system of the environmental equipment (heater-air conditioner) or by separate system(s); such as power intake, exhaust ventilator(s), and static roof vent(s).

3.13.7 Environmental controls. Adjustable, manual, or thermostatically operative controls shall permit heating and/or air conditioning and ventilating in either compartment, without affecting the other compartment or simultaneously heating one compartment and air conditioning the other. Switches and controls shall be located in panels and be readily available and identified for function and operating position (see 3.8.7 thru 3.8.7.2). Switches and control components shall exceed in capacity the amperage and resistance requirements of the motors. Blower or fan motors shall be at least three speed design. Shut-off valves for the heating (hot water) system(s) of each compartment shall be provided to allow servicing and shall permit the use of the balance of the unaffected system(s). Air systems shall have adjustable louvers to direct the flow of air.

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3.14 Communications.

3.14.1 Communication equipment. All ambulances shall be equipped with a 12 volt DC two-way (mobile) radio, intercom, public address, and electronic siren. Telemetry equipment shall be optional and specified as required. Switches and controls shall be panel mounted with switch positions identified.

3.14.2 Two-way (mobile) radio. Unless specified (see 6.2), that the consignee shall furnish radio equipment, the supplier shall furnish and install ready-to-use, two-way radio as specified herein (3.14.4) and coordinate installation with consignee to insure compatability. All ambulances shall be provided with sufficient space for the radio, convenience features (antenna openings, wiring, panels, etc.), that are required for the installation and servicing of such radios.

3.14.3 Antenna, cable and mountings. Antenna shall meet the requirements of EIA Standard RS329 Part II. The coaxial lead-in wire furnished shall be of the impedance, loss, and power handling capability as specified by the radio manufacturer. When it is specified that a two-way radio and antenna will be installed by the purchaser, the supplier shall provide a plate and opening, and coaxial lead-in wire from the ventilated radio storage compartment to the antenna port.

3.14.4 Two-way (mobile) radio equipment. A solid state (no tubes) mobile two-way radio shall be furnished, with all normal accessories. This radio shall meet the requirements of EIA Standards RS152B (Transmitters), RS204A (Receivers), RS329 Part II (mobile antennas) and, if applicable RS220 (Continuous Tone Coded Squelch Systems RS374 (Selective Signalling). The transmitter power output shall be *(as specified) watts minimum. Microphones shall be provided for transmitting at both the driver's position and in the patient's compartment. Receiver audio power output shall be *(as specified) watts minimum. Selectable speaker outputs, singly and in combination shall be provided at the driver's position, patient's compartment, and through the public address system. Radio interference suppression see 3.7.10. *Power to be specified (see 6.2) consistent with local emergency communication requirements.

3.14.5 Intercom system. An inter-communication facility shall be provided between the driver and patient compartments. The intercom shall be solid-state, independent of the two-way (mobile) radio equipment; have speaker/microphone unit in the patient compartment in stand-by "talk" mode; located near technician's seat, and any necessary talk/listen switching controlled by the driver.

3.14.6 Siren - public address system. The combined electronic siren and public address system furnished shall include upgrading features; radio amplification capability, complete with two speakers (see 3.14.7); noise cancelling microphone, manual and automatic siren undulation, and yelp; plug-in connections and connecting cables, foot switch; and all normally furnished accessories. In siren operation, the power output shall be 100 watts, minimum. In voice operation, the power output shall be 45 watts minimum through exterior mounted speakers. The public address amplifier shall be independent of the two-way (mobile) radio.

3.14.7 Speakers. Two (siren/public address/radio) speakers shall be installed outside on the vehicle cab. The speakers shall be chrome-plated, streamlined, weatherproof, on reinforced mounting to prevent drumming and vibration. Speakers shall be mounted in the following manner: Both speakers facing forward at 15 degrees off the center line of cab roof. Speakers shall be connected in phase to prevent noise cancellation. Speaker location shall permit servicing of the vehicle without removing the speakers or interfering with the intent or vision of the warning lights.

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3.15 Additional systems, equipment, accessories, and supplies.

3.15.1 Additional and optional equipment. When specified, the additional or optional system(s), equipment, accessories, and supplies (selection follows) shall be in addition to the standard ambulance component systems and devices specified herein. These shall be selected by the purchaser to meet the users needs. Optional items may be further described in authoritative documents. In no event shall the specified or furnished optional item(s) supersede or reduce the quality and intent of the ambulance, but shall enhance its design and purpose. The materials, devices, items, and fabrication if not specifically described shall be not less in quality, strength, performance, and service, than those normally provided by the most reputable manufacturers.

3.15.2 Standard mandatory miscellaneous equipment. Unless otherwise precluded elsewhere in this specification or contract, each ambulance shall be equipped with, but not limited to the following:

a. Fire extinguisher; dry chemical minimum 5 lb. unit, quick-release bracket mounted in patient compartment.
b. Backboards; wood or aluminum, long minimum size 72 x 18 inches, short minimum size 32 x 18 inches with straps.
c. "No Smoking" signs conspicuously placed in cab and patient compartments, warning of oxygen use dangers.
d. Overhead grab rail; minimum 60 inches long maximum 3 inch depth, on ceiling over the primary patient.
e. A hinged, half-ring lower-extremity splint with a minimum ring size of 9 inches and minimum over-all length of 43 inches, with padded ankle-hitch or skin-traction device, with either a built-in turnbuckle device or a supplemental "Spanish windlass" device.

3.15.3 Optional equipment. When specified (see 6.2), in addition, the ambulance shall be equipped with, but not limited to the following items:

1. Battery charger; with electric "T" polarity plug-in outlet to 110 volt receptacle.
2. Tachograph; 90 MPH, with 24-hour chart, plus light and siren recording.
3. Intercom; two-way visible intercommunication system between driver and technician with amber, red and green lights.
4. Mud-guards; to protect lower front-end of body (for type I).
5. Back-up alert; audible warning device activates when vehicle is shifted into reverse.
6. Drapes; fire-retardant cloth drapes for each window of patient compartment except partition; installed on polished metal drapery rods.
7. Drug compartment; security compartment, minimum 8" x 8" x 12" long; with hidden hinged door with key lock in patient compartment.
8. Power winch, electric; front center mounted, minimum 3 ton capacity, with wire rope, chain, hook, and roller guides.
9. Rear door medicine compartment; recessed in lower rear door(s) (type I and III).
10. High-intensity cot light; with flexible or adjustable shaft located near head of primary cot.
11. Patient fan; located near primary cot, minimum 250 CFM rating.
12. Rear step stowage compartment, (types I and III).
13. Partition: with access door between compartments for types II and III. (see 3.10.2).
14. Window; clear, right side patient compartment, minimum size 16" x 36" long.
15. 3,000 watts, 110 volt AC power (see 3.7.8) additional AC power for rescue and emergency equipment.
16. Power plant heaters, minimum -30°F. startability (see 3.6.3.2.1, severe cold area only).
17. High altitude operations, engine emissions adjustments and controls (see 3.6.4.3).
18 Extrication equipment, as specified (see 3.11.2).

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3.15.4 Medical, surgical, and bio-medical equipment. When specified (see 6.2), the ambulance shall be equipped with, but not limited to the following:

M1 - Rail system; for attaching various medical devices to the rail with quick release guides; rail minimum 90" long, mounted near or at primary cot; for example, Modura (Fairfield) Rail System or equal.
M2 - Rail system guides; minimum 5 for M1 rail for mounting additional equipment.
M3 - Stop clock; manual wound; with separate holder.
M4 - Suction jar for patient aspiration with disposable inserts or spare bottle, with holder.
M5 - Sphygmomanometer; with compression bag, cuff, stethoscope, inflating bulb and mounting.
M6 - Ventilators (2) portable, squeeze bag valve mask units with face holders, etc. (see 3.12.2.1).
M7 - Humidifier and oxygen flowmeter; with holder (in addition to 3.12.1 and 3.12.2).
M8 - Pulse tachometer, battery-operated; visual and audible type; with holder.
M9 - Ceiling IV holders/hooks, recessed or near flush, in addition to cot IV holders (see 3.11.9).

M10 - Emergency medical care kit; 24 units contained in dust and moisture proof steel case.
M11 - Scoop stretcher, folding type.
M12 - Scoop stretcher, non folding type.
M13 - Style 3 cot (3.11.5) floor holding and securing devices (state location).
M14 - Stretcher, combination stretcher wheel chair, in lieu of standard stairway stretcher (see 3.11.5).

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3.16 Preparation for painting, color, and markings.

3.16.1 Preparation for painting. Ambulance body and all attached equipment exterior surfaces, except polished metal parts, shall be thoroughly cleaned, treated, and coated with a firm primer and preservatives with rust inhibiting properties, and painted in the finish color as specified. Ferrous metal interior surfaces shall be painted or, when not exposed for painting, shall be treated or coated to resist corrosion. Chassis and chassis frame components shall be preserved and finished in accordance to industries standard practice.

3.16.2 Salt spray resistance. Treated exterior sheet metal of the ambulance body shall be capable of withstanding 250 hours of salt spray tested in accordance with ASTM B 117-64. The specimen used for the salt spray test shall be run through all steps of the cleaning and treating process, including priming. The primed specimen shall be scored from corner to corner using a sharp knife. After the test, the specimen panels shall exhibit no failure or rusting, including not more than 1/8- inch rust or blister creepage from the scored lines.

3.16.3 Color, paint, and finish. The exterior color of the ambulance shall be basically white in combination with orange strip and blue lettering and emblems. The exterior finish shall be an acrylic composition paint. Final color coats of paint shall have a film not less than 1.25 mils. thick. The band (stripe) of orange not less than 6 inches wide, nor more than 14 inches wide shall encircle the entire ambulance body configuration at the belt line (below the lowest edge of cab windows). This band when viewed horizontally shall appear as a stripe near parallel to the road. The interior finish shall be the manufacturer's standard light color harmonizing with the color of upholstery. The final film of painted surfaces shall be smooth and uniform, free of grit, streaks, blushing, runs, sagging, blisters, pin holes, or other irregularities of surface. Exterior finish paint shall not be required on the underbody and inside surfaces of the body skirting.

3.16.3.1 Color parameters. The exterior surface including the wheels shall be manufacturer's standard gloss white. The orange strip (band) and blue markings shall comply to the Munsell System and its visual attributes of hue, value, and chroma as described in ASTM method D1535 (American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103). These colors identify in Munsell notations as follows:

Orange, 5 OYR 6.0/15 with tolerances in hue, value, and chroma of:

Hue +, 6.25 YR; Hue -, 3.75 YR; Value + 6.5;
Value -, 5.5; Chroma +, unlimited; and Chroma -, 13.

Blue, 2.5PB 3.5/10 with tolerance in hue, value, and chroma of:

Hue + 4.5 PB; Hue -, 10.0 B; Value + 4.0;
Value -, 3.0; Chroma +, unlimited; and Chroma -, 8.

(Color chips for these color standards are available from the National Technical Information Service (NTIS) Springfield, Va. 22151).

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3.16.4 Emblems, and markings. The material for the emblems and markings shall be applied using reflectorized material conforming to Federal Specification L-S-300, type I, class 3 reflectivity 1. The reflective color used shall be blue (color a). The emblems and markings shall be of the type, size, color and location as follows:

A. Front markings
   a. The word "AMBULANCE" in block, blue letters, not less than 4 inches high, shall be mirror image centered above the grill.

   b. Block type blue, "Star of life" conforming to figure 4, shall be not less than 3 inches, on a 4 inch white field, (size A) located both to the right and left of the word "AMBULANCE".

B. Side and rear markings
   a. The word "AMBULANCE" shall be in block blue letters on the white field, of not less than 6 inches in height, centered, along side, or under the "Star of life" (b) on each side and rear of the vehicle body.

   b. A block type blue "Star of life", conforming to figure 4 (size B) of not less than 12 inches on a white field. The "Star of life" emblems may be provided on each window glass of the rear doors, when white field space is limited.

C. Top markings
   a. Block type blue "Star of life", conforming to figure 4, (maybe without the white Staff of Aesculapius), of not less than 16 inches, (size C) shall be conspicuously centered between the warning lights on the roof top.

3.17 Undercoating. Ambulance shall be undercoated in accordance with Mil. Std. 1223 unless rustproofed (3.18).

3.18 Rustproofing. When specified (see 6.2), ambulance shall be rustproofed in accordance with Fed. Std. No. 297. (Undercoating will not be required when vehicle is rustproofed).

3.19 Data and caution plates. Data, caution, and identification plates or decals listing data specifically applicable to the vehicle and the furnished equipment shall be permanently and conspicuously installed on the vehicle. No other trademark(s), names, or other identification, other than authoritatively specified and the vehicle manufacturers name shall be applied to the vehicle. Data plates shall be of nonferrous material, except for instruction plates or decals mounted inside the vehicle, information required for data plates shall be inscribed or stamped in a manner that the legend will be discernable if inadvertently painted over.

3.20 Manuals and handbooks of instructions. The supplier shall furnish one copy of the vehicles operating instruction handbook, repair manual, and parts book with each vehicle supplied under this specification. Also, an instruction handbook(s) shall be furnished for all ambulance related accessory components, equipment, systems furnished as part or parcel of the emergency medical care vehicle. The handbook(s) shall contain all installation and operating instructions, drawings, illustrations, manufacturers part numbers, lubricating instructions, assembly and disassembly instructions, and safety precautions to insure proper installation, operation, and maintenance. Wiring diagrams requirements, see 3.7.2.

3.21 Servicing and adjusting. Prior to acceptance of the ambulance by the purchaser the supplier shall service and adjust the vehicle in accordance with GSA Form 1455, and the operational checks (4.9.1), and shall perform the following; focusing of headlights; alignment of front wheels and steering wheel; truing all tires or replaced with true tires; complete vehicle lubrication service with grades of lubricants recommended for the ambient air temperature at the delivery point; and elimination of vehicle rattles, noises, and squeaks.

3.22 Special requirements. Specify (see 6.2) unusual operating conditions, miscellaneous items, exceptions, markings, end use, etc.; not specified herein. If in conflict with the manufacturer's standards and options, those specified shall take precedence.

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3.23 Workmanship. Defective components shall not be furnished. Parts, equipment, and assemblies, which have been repaired or modified to overcome deficiencies shall not be furnished without the approval of the contracting officer. Welded, bolted, and riveted construction utilized shall be in accordance with the accepted standards of industry. Component parts and units shall be manufactured to definite standard dimensions with proper fits, clearances and uniformity. General appearance of the vehicle shall show no evidence of poor workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the purchaser. The purchaser reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified are classified as follows:

(a) First production ambulance inspection and test (see 4.3 and 4.4).
(b) Purchaser verification (see 4.8).
(c) Quality conformance inspection (see 4.9).

4.3 First production ambulance inspection. Unless otherwise specified (see 6.2), the first production ambulance of each type specified under contract shall be inspected in accordance with the examination in 4.3.2 and all tests under 4.4. The first ambulance produced under the contract shall be completely representative of the balance of the vehicles to be delivered under the contract. The purpose of the inspection shall be to determine vehicle conformity with the contract. The ambulance shall be inspected by the contractor at his plant under the direction and in the presence of purchasers representative(s). Inspections shall be conducted during normal daytime working hours. The cost of inspections, examination, all tests, and vehicle support, exclusive of personal expenses incurred by participating purchasers representatives shall be borne by the contractor. Acceptance of the first production vehicle shall not constitute a waiver by the purchaser of its rights under the provisions of the contract.

4.3.1 Vehicle weights. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight and the payload as specified in 3.5 thru 3.5.4. Calculated imposed loads on front and rear axle will be utilized to ascertain that suspension, axles and tires furnished are of adequate capacity to meet contract requirements.

4.3.2 Examination. First produced ambulance shall be visually examined to determine compliance with the requirements in section 3.

4.4 Tests

4.4.1 Test criteria. The ambulance shall be prepared for operation in accordance to manufacturer's recommendations, and 3.21, manned and loaded to simulate maximum payload conforming to type, class, and plan specified. Road test course shall be approved by the procuring activity, or purchaser representative. Substitution of new vehicle or replacement of major component(s) may require complete retest at the discretion of the purchasers representatives. If the ambulance manufacturer has previously tested a similar model vehicle with identical components, a certification for the specified test data verifying conformance to this specification may be provided in lieu of the first production ambulance tests. Certification(s) and previous test results, if available, shall be forwarded to the procuring activity along with the vehicles engineering data prior to purchase or bid opening.

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4.4.2 Performance tests. The ambulance and equipment as furnished, shall be tested to verify conformance of the following:

(a) Vehicular operation in temperature conditions (see 3.4.2).
(b) Vehicle physical dimensions (see 3.4.11 thru 3.4.11.6).
(c) Cooling system (see 3.6.4.5).
(d) Tire changing tools (see 3.6.13).
(e) 110 volt AC electrical power system (see 3.7.8).
(f) Warning lights (see 3.8.2).
(g) Flood light (see 3.8.3).
(h) Spotlight(s) (see 3.8.4).
(i) Body, static load test (see 3.10.5).
(j) Oxygen, main supply and installation, leak and pressure test (see 3.12 and 4.4.5).
(k) Suction aspirator system, performance test (see 3.12.3).
(l) Heating system(s) (see 3.13.4).
(m) Air conditioning system(s) (see 3.13.5).
(n) Ventilation (see 3.13.6).
(o) Environmental controls (see 3.13.7).
(p) Two-way radio and controls (see 3.14.4).
(q) Salt spray resistance (see 3.16.2).
(r) Road test (see 4.4.3).
(s) Water spray test (see 4.4.4).

4.4.3 Road test. The vehicle shall be subjected to a minimum 150 mile road test of which 75 miles shall be continuous miles on paved highways at highway speeds up to 70 mph; 30 miles on city streets; 15 miles on gravel or dirt roads at speeds up to at least 35 mph; and not less than 5 miles in simulated or actual cross-country operation at speeds applicable to the terrain. Cross-country operation is defined as travel over open fields, rolling and side-sloping hills, rough and muddy terrain. Class 2 vehicles shall demonstrate cross-country operation in 4-wheel drive for an additional 20 miles. Ambulances shall meet performance requirements specified in 3.4.4 thru 3.4.10 during road tests or by certification. Balance of the 150 miles road test may be accumulated during other tests and checks requiring vehicle movements. After completion of the road test, vehicle shall be subject to the water spray test.

4.4.4 Water spray test. The ambulance shall be subjected to a water spray test for approximately 15 minutes. The spray shall be delivered by nozzles operating at 25 pounds per square inch (psi) water pressure, sufficient in number and placed (approximately 3 feet from the body) to afford full coverage of sides, roof, front, rear and under carriage of the vehicle. Ambulance undergoing quality conformance inspection (4.9) shall be subjected to water spray test for not less than 2 minutes duration. Evidence of water leakage shall be cause for rejection until leaks are corrected.

4.4.5 Oxygen system tests. The installed medical oxygen system and components (see 3.12.1) shall be leak tested at 150 percent of operating pressure, for a time period of four hours. The system shall be tested with water nitrogen gas or equal and kept decontaminated. After the successful completion of tests, the system shall be capped, then tagged with date and signature of person and firm performing the tests.

4.5 Inspection failure of first ambulance. Failure of first production ambulance to meet requirements and successfully complete examinations and tests shall be cause for nonacceptance of any or the contract quantity, until deficiencies and evidence of the corrective action precludes recurrence of similar deficiencies. Failure of the vehicle to successfully complete test shall not constitute an excusable delay in meeting scheduled deliveries.

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4.6 Test failure of first ambulance. Vehicle utilized for the road test shall successfully complete 150 miles of test. Rejection of the test vehicle shall be for deficiencies, including but not limited to the following:

(a) Damage caused by collision.
(b) Failure of any major component.
(c) Vibration due to misalignment of wheels, frame, driveshaft, etc.
(d) Vibration due to type of body construction or mounting.
(e) Evidence of abnormal tire wear due to misalignment or unbalanced wheels/tires.
(f) Failure of any vehicular safety device such as brakes, steering assembly, windshield washers and wipers, or electrical circuits.
(g) Evidence of structural weakness in any part of the vehicle, vehicle components, or accessories.
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation
(i) Failure of any vehicular performance requirements.

4.7 Production sample. Upon acceptance of the first production ambulance, it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract. Vehicle shall be then cleaned, serviced, and refurbished to the extent required for delivery of a new vehicle to the purchaser. (Single unit purchasers see 6.2,n).

4.8 Purchaser verification. Quality assurance operations performed by the supplier will be subject to purchaser verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the supplier's inspection are being properly applied. Failure of the supplier to promptly correct observed deficiencies shall be cause for suspension of acceptance until conformance to specification criteria has been demonstrated.

4.9 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to all production vehicles offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (e) as follows, and failure of any unit to pass an examination or test shall be cause to refuse acceptance of the vehicle until corrected.

(a) Production methods verification (see 4.8).
(b) Workmanship inspection (see 3.23).
(c) Ambulance operational checks (see 4.9.1).
(d) Water spray test (see 4.4.4).
(e) Preparation for delivery examination (see 4.9.2).

4.9.1 Operation checks. Operational checks of the ambulance shall cover all controls, electrical systems and devices, doors, windows, cabinets, accessories, in and outside the ambulance. Ambulance shall be driven at all speeds, turns made at minimum radii, brakes tested for dependability, checked for rattles and squeaks, and compliance to 3.21. All controls, and mechanisms shall function and operate as intended.

4.9.2 Preparation for delivery examination. Preservation, packaging, and marking for shipment as applicable shall be examined to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preparation. Unless otherwise specified (see 6.2), the ambulance(s) shall be preserved and packaged for mobile delivery in accordance with the supplier's standard commercial practice, insuring carrier acceptance and safe delivery to destination in compliance with regulations applicable to the mode of transportation.

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6. NOTES.

6.1 Intended use. The vehicles covered by this specification are intended for on and off the road emergency medical services. The ambulance is designed with a driver compartment or cab, a patient compartment accommodating one medical technician, and two litter patients. The ambulance (properly equipped and supplied) is intended to provide optimal emergency care, light extrication work at the scene of emergency, and intensive life support for at least one patient during transit.

6.1.1 Federal specification coverage. Federal specifications do not include all the varieties of ambulances as indicated by the title of the specification, or which are commercially available. This specification covers only the ambulances generally purchased, or used to provide service under contract, or approved for funding by the Federal Government (see 1.3).

6.1.2 Precautions and observations. Due to the variety of ambulance models covered in this specification, purchasers should be aware that some items of equipment, components, etc., described herein are not necessarily compatible with all vehicles. It should be ascertained before specifying (ordering) that ambulance manufacturers offer such equipment, devices, or options, (reference manufacturer's data books,brochures) for the type and class ambulance desired.

6.1.3 Definition of Government-purchaser. Government or purchaser as used in the context of this document means; the Federal, State, or political subdivision Government, or any purchaser who cites this specification.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement document.

(a) Title, number, and date of this specification, and amendment number if any.
(b) Quantity, type, and class ambulance required (see 1.2.1).
(c) Patient compartment floor plan (plan A is standard); specify if plan B is desired (see 3.1.5).
(d) State if California Emission Controls and test required (see 3.6.4.3).
(e) Manual four speed transmission, for class 2 (4 x 4) and transfer case, automatic transmission is standard, detail other powertrain requirements. Check suppliers option availability. (see 3.6.5.2.1 and 3.6.5.3).
(f) Axle ratios, state if operating in mountainous terrain, (do not specify ratios) (see 3.6.5.6).
(g) Wheels, specify if dual rear wheels are specifically desired for types I or III vehicle (see 3.6.7).
(h) Tires, specify where snow tires instead of regular tires desired (see 3.6.8 and 3.6.9).
(i) Warning lights, (blue and white colored lights are standard),specify if other colors are required by state or local regulations (see 3.8.2).
(j) Two-way (mobile) radio communications, specify if purchaser will furnish radio and components to contractor, and cite space and pertinent information. If contractor is providing communication equipment as specified, cite brand(s), type of power outputs, frequencies etc., desired and complete the details for radio installation and provide a coordinator's name, address and telephone number (see 3.14 thru 3.14.7).
(k) Additional systems, equipment, accessories, and supplies (see 3.15 thru 3.15.4), cite additional parameters where necessary:

    a) Select optional (vehicle) equipment desired, 1 thru 18 (see 3.15.3).
    b) Select medical (ambulance) equipment desired M1 thru M14 (see 3.15.4).

(l) Rustproofing, if required, undercoating is standard (see 3.18).
(m) Special requirements, state additions changes or deletions to the specification (see 3.22).
(n) Inspection, quality assurance provisions (see section 4).
    (It is recommended that "Ambulances" procured under this Specification be inspected at the suppliers plant, fully complying to at least (c) of 4.2 Quality conformance inspection (see 4.9) for quality assurance).

    1 - State which classification(s) of inspection desired (see 4.2).
    2 - First production ambulance inspection, cite any differences (see 4.3 and 4.7). Recommended for volume procurement only.
    3 - Quality conformance inspection, cite any differences (see 4.9)

(o) Preparation for delivery, other than specified (see 5.1). State mode of delivery prefered.
(p) Procurement requirements (see 6.3).

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6.3 Procurement requirements. When specified (see 6.2) invitation for bids, contract or orders will contain the following contractual requirements (see 6.4 thru 6.7), except in those instances where it is determined that inclusion thereof would not be to the best interest of the Government or purchaser.

6.4 Warranty. The manufacturer or contractor shall warrant the entire ambulance, furnished equipment, items and devices, (except 6.4.1), against parts failure or malfunction due to design construction, or installation, errors in assembly of components, defective material and workmanship, for a minimum period of 12 months or 12,000 miles of operation, exclusive of authorized accumulated driveaway mileage, whichever may occur first. The vehicle chassis, chassis component (drive line etc.) and applicable body or cab may be warranted in accord with their standard vehicle warranty policy.

6.4.1 Warranty exceptions. Unless caused by defective material and workmanship, the manufacturer or contractor need not assume liability or reimburse the Government regardless of vehicle age and mileage. The following may be considered warranty exceptions.

(a) Abuse, negligence, or alteration of original parts or adjustments.
(b) Engine tune-ups.
(c) Brake adjustments.
(d) General tightening, head lamp adjustments.
(e) Wheel alignment.
(f) Tires, tubes, and batteries (if warranted by their manufacturers).
(g) Miscellaneous expense such as gasoline, towing, telephone, traveling, lodging, loss of personal property, or sales tax.

6.5 Illustrations and drawings. When specified in the invitation for bids (see 6.2), the bidder shall furnish with his bid two sets of illustrations and outline drawings with dimensions to assist the purchasing and using agencies in determining whether the vehicles offered are adequate to perform the specified services and requirements.

6.6 Repair parts and service. As a continuous operation of the vehicles contemplated by this specification is of utmost importance, it is necessary that the successful bidder be in a position to render prompt service and to furnish replacement parts. Accordingly, bidders shall indicate the extent of their ability to render prompt service by furnishing a list of branch offices or agencies where complete stocks of repair parts are maintained and can be secured within a reasonable time after ordering by part number from the manufacturer's part book and at such discount as may be quoted from year to year by the manufacturer of the vehicle purchased under this specification.

6.7 Statement of Origin or bill of sale. A manufacturer's statement of Origin or Bill of Sale showing the applicable purchase order number is required for each vehicle procured under this specification. Unless otherwise specified, such documents shall be forwarded to the consignee.

6.8 Oxygen tanks. Following are the approximate dimensions, weights, and capacities of the various size oxygen cylinders referenced in this specification:

	TANK	
	D	M
Outside diameter (inches)	4 1/2	7
Overall heights, without valve (inches)	20	47
Capacity (gallons)	95	800
Capacity (liters)	360	3000
Capacity, flow @3 to 10# (minutes)	40	325
Capacity (cubic feet)	13	106
Weight, empty (pounds)	13	75
Weight, full (pounds)	14	83

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6.9 Star of Life. The "Star of Life (figure 4) is a six-barred cross upon which is superimposed the Staff of Aesculapius (es "cu-la" pi-us) who, in Roman Mythology, was the god of medicine and healing. Following are the dimensions referenced in this specification:

Dimensions:	Size A	Size B	Size C
Length of bar	3"	12"	16"
Width of bar	3/4"	3"	4"
Length of Staff	2 1/2"	9 1/2"	12 1/2"
White background (if required)	4" square	14" square	18" square

All angles 60°
Deviations shall be proportionate.

MILITARY INTERESTS:
Army - AT
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITIES
DOT
HEW
VA

PREPARING ACTIVITY: GSA-FSS-ASD

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price 40 cents each.

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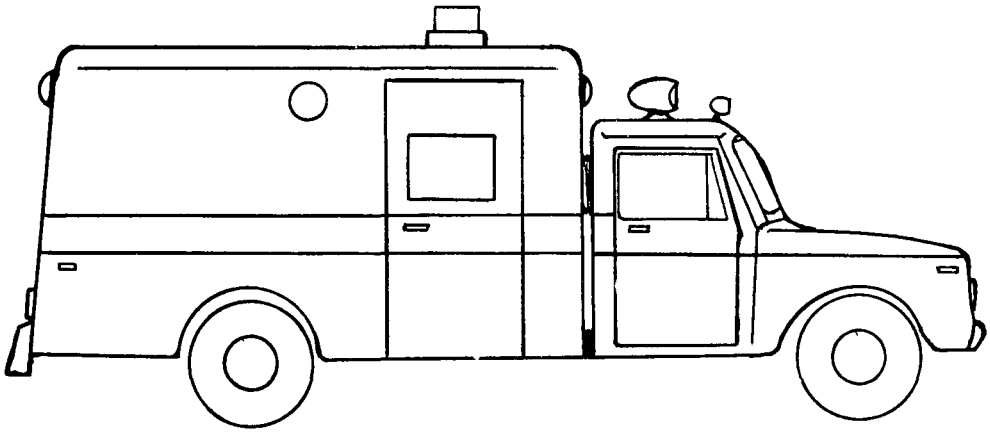


FIGURE 1. Type I Ambulance

(Conventional cab-chassis with modular ambulance body)

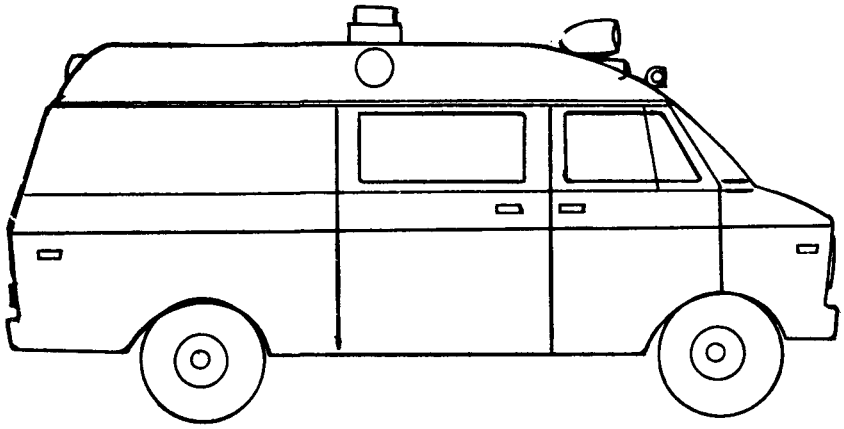


FIGURE 2. Type II Ambulance

(Standard Van, forward control, integral cab-body ambulance)

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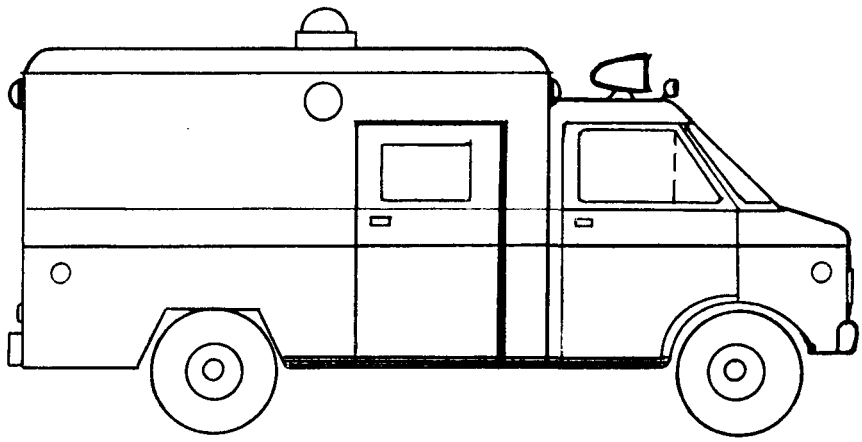


FIGURE 3. Type III Ambulance

(Specialty Van, forward control integral cab-body ambulance)

"STAR OF LIFE SYMBOL"

Emergency Medical Care Vehicle-Ambulance

The "Star of Life" is a six-barred cross upon which is superimposed the Staff of Aesculapius (es"cu-lā'pĭ-us) who, in Roman Mythology, was the god of medicine and healing.

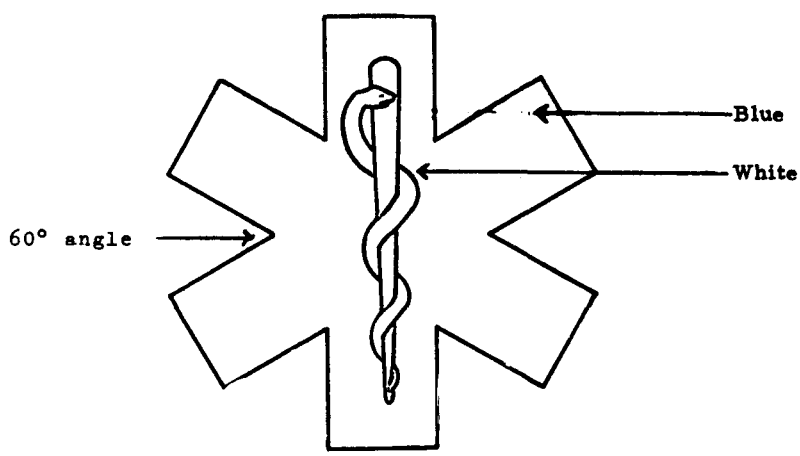


FIGURE 4.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: General Services Admin., FSS (FMA), Wash., DC 20406

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

KKK-A-1822: AMBULANCE
Emergency Medical Care Vehicle

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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